

Unit-wise Questions

Unit 1: Cloud Computing Fundamentals

1. What are cloud deployment models? Describe private, public, and hybrid clouds. Discuss their advantages and disadvantages.
2. Define cloud computing. How does it differ from traditional computing?
3. Explain the evolution of cloud computing. What are the benefits and limitations of cloud computing?
4. What are the different models for deployment in cloud computing? Explain any one of them. Discuss Infrastructure as a Service (IaaS) vs Software as a Service (SaaS) vs Platform as a Service (PaaS).
5. Describe the cloud service models SaaS, IaaS, and PaaS. Provide examples of each and explain their use cases.
6. Compare and contrast cloud computing with grid computing.
7. What are the economic benefits of cloud computing for businesses?
8. What do you mean by cloud and cloud computing? Give a brief note on the advantages and disadvantages of cloud computing.
9. Public cloud is less secure. Justify.

Unit 2: Cloud Computing Service Models

1. Describe the various cloud computing service models. Discuss the key characteristics and applications of SaaS, IaaS, and PaaS.
2. Explain the concept of a hybrid cloud. What are its advantages and disadvantages?
3. What is the significance of cloud-based big data/real-time analytics? How does it benefit businesses?
4. Explain the role of load balancing in cloud computing. How does it improve performance?
5. What is cloud architecture? Explain its components and how it supports cloud services.
6. Explain the role of SOA (Service-Oriented Architecture) in cloud computing. How does it improve performance through load balancing?
7. Discuss the process and benefits of deploying a web service from inside and outside of a cloud.
8. What is the role of hypervisor in virtualization? Briefly explain the different types of hypervisors with a neat diagram.
9. What is the role of hypervisors in virtualization? Provide examples of different types of hypervisors.

Unit 3: Service Management

1. What are Service Level Agreements (SLAs) in cloud computing? Discuss their importance and key components.
2. Describe the authentication process in cloud computing. What are the requirements for client access?
3. What is Amazon Web Services (AWS)? Explain the installation and configuration process of AWS. Discuss its main features.
4. What is Microsoft Azure? Define Azure services and explain them with the help of suitable examples.
5. Define cloud architecture. How does it support scalability and reliability in cloud services?
6. Explain the importance of data scalability in cloud services.
7. Describe the main features and benefits of Amazon EC2.
8. What is cloud contracting? Discuss its commercial and business considerations.

Unit 4: Security Concepts

1. What are the various cloud security approaches and cloud security challenges? Discuss briefly.
2. Discuss the challenges and solutions for cloud security. How can encryption and tokenization enhance cloud security?
3. Explain cloud security alliance standards. How do they help in securing cloud environments?
4. What are the common security challenges faced by cloud computing? How can they be mitigated?
5. Discuss cloud security models and related patterns.
6. Explain the role of VMware in server virtualization.
7. What are the objectives and benefits of using VMware in server virtualization?
8. Describe the authentication process in cloud computing. What are the requirements for client access?

Unit 5: Case Studies and Presentations

1. Write short notes on any TWO:
 - Amazon EC2
 - Private Cloud
 - Big Data Analysis

- Google App Engine
- Cloud-based Big Data/Real-time Analytics
- Commercial and Business Considerations in Cloud Computing
- Mainstream cloud security offerings
- Virtualization for enterprise
- Installation of AWS
- Cloud computing benefits for SMEs
- Storage as a Service (StaaS)
- Improving performance through load balancing
- Eucalyptus
- Database and data stores in the cloud